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Final Research Paper

Effect of Cigarette Price, Annual Income, Age, and Years of Schooling on the Consumption of Cigarettes

I. Introduction

It is almost universally well known in today's society that smoking cigarettes highly contributes to the likelihood of lung cancer and early death. However, the addictive quality of nicotine makes this habit a difficult one to break. One way that the government has tried to curb this public health issue is to raise the prices of tobacco exponentially in an effort to discourage buyers. However, people continue to smoke. In order to understand how end smoking on a large scale, it will be important to understand the factors that influence certain demographics of people to smoke cigarettes. The objective of our group's study is to discover the relationship (correlation) between the consumption of cigarettes and cigarette prices and other variables such as age, income, and education. Each of these categories is expected to have varying amounts of habitual smokers at different levels. This leads us to the research question: "How does cigarette prices, annual income, age, and years of schooling affect the cigarette consumption of an individual?" Our study will attempt to find the variable with the most impact on these relationships, and the 807 observation sample set that we analyzed will help us do so. Increasing public awareness about the health consequences of smoking and countless efforts by many different institutions and organizations to reduce smoking have influenced other studies to be performed similar to this one. Policy makers need the type of data that we will be analyzing to be able to better determine the most influential variable that affects cigarette consumption. In doing so, these policy makers so that can form strategies to target specific variables in order to reduce smoking in the general population. It is also important to keep in mind that the data analyzed from the observations in this research comes from a sample subset of a larger population, and may not accurately reflect the adult population at large.

Our results were interesting regarding education level and the presence of state regulations prohibiting smoking in restaurants. Education and regulations were both found to decrease the consumption of cigarettes substantially. Creating a dummy variable for high school graduates vs. non graduates also generated interesting data regarding the decrease of consumption with an increase in education, which can be referred to in the data analysis section.

II. Literature Review

Other studies have been done by agencies and companies concerned with public health in relation to tobacco use. Increasing cigarette prices in the form of excise taxes has been one of the government's most utilized tactics to discourage cigarette consumption. As expected, higher prices successfully discourage a significant proportion of people from starting to smoke. Another identifiable variable that affects cigarette consumption is age. Young adults have been found to have a more inelastic demand for cigarettes when it comes to price, because they do not typically have steady or considerable incomes.

In "*Price, Tobacco Control Policies and Smoking Among Young Adults,"* Chaloupka and Wechsler state that "in the face of the anti- smoking campaign, cigarette smoking remains stubbornly high, particularly among youths and young adults" (p. 3). This statement highlights the prevalence of smoking despite public service announcements and campaigns to spread awareness about the dangers of tobacco use. The fact that smoking prevails despite higher taxes and awareness confirms the importance of studying other variables in the demographic of people who are choosing to smoke. Chaloupka and Wechsler's study focused specifically on college students, which is an interesting demographic to identify because it is both a specific age and education level, two of the variables we study in our paper. After performing the appropriate

statistical analyses, Chaloupka found that "the average overall estimated price elasticity of demand for college students of -1.43 is more than three times the consensus estimate of -0.4 for adults. This suggest that increases in cigarette excise taxes, which would increase cigarette prices, are a very effective means of reducing cigarette smoking among youths and young adults" (p. 25). The study also produced results to indicate that higher taxes would reduce smoking participation and average cigarette consumption among U.S. college students. Although our study does not focus on college students alone, this demographic of people may be the most important in successfully and permanently reducing cigarette consumption in subsequent generations. The current generation of young people is the one who will raise the next generation of children, so it is especially beneficial to discourage them from smoking.

Links have also been found between lower education level and higher instances of cigarette consumption. In another literature source "*Cigarette Smoking and Lung Obstruction Among Adults Aged 40-79,"* the authors found that"among adults aged 40-49, the percentage of adults who smoked declined with increasing levels of education" (p. 5). This could be attributed to young adults spending less time in a controlled environment where smoking is prohibited and frowned upon. It could also be a factor because fewer years of schooling may contribute to a greater lack of understanding about the harsh impact of tobacco on the body in a more scientific, chemical sense. It is important to address that people with fewer years of education consume more cigarettes on average, despite their average lower salary. This counterintuitive point is part of a larger socially constructed issue that will be hard to solve with the data at hand, or maybe impossible to figure out with statistics alone. This paper also talks about race in relation to cigarette consumption. Although our only variables for race were black and white (this data set

also included people of Hispanic origin) it is important to recognize that race is a socially significant variable that very well could have an impact on cigarette consumption.

III. Data description

Our group used the Woolridge data set 'smoke.dta' which has 807 individual observations; each with an array of variables. This dataset can be accessed using the Stata command **bcuse smoke**. In our study we will be focusing on 6 variables: years of schooling (educ), state cigarette price (cigpric), age in years (age), annual income in US dollars (income), cigarettes smoked per day (cigs), and state restaurant smoking restriction (restaurn). We also added a dummy variable called "hsgrad." This variable has a value of 1 if a person has 12 or more years of schooling, and value 0 otherwise.

For our study, the primary variable of interest is consumption of cigarettes at the individual level. In Stata, this variable is called 'cigs' and is measured in cigarettes smoked per day by each observation.

For our secondary dummy variable our group is using state restaurant smoking restriction for the state the observation lives in. This variable is called 'restaurn' in Stata and assumes value =1 if the state that the observation lives in has smoking restrictions in its' restaurants and =0 if the state that the observation lives in does not have smoking restrictions in its' restaurants.

For our secondary continuous variable we have education, cigarette price, age, and income. Education is referred to as 'educ' in Stata and is measured in years of schooling. Cigarette price is referred to 'cigpric' in Stata and is measured in US dollars of state cigarette price where the observation resides in. Age is referred to as 'age' in Stata and is measured in

years of age. Income is referred to as ‘income’ in Stata and is measured in US dollars in annual wage of the observation.

The mean for cigarette consumption in all observations is 8.69 cigarettes per day smoked with a standard deviation of 13.72 with a minimum value of 0 cigarettes smoked per day and a maximum value of 80 cigarettes smoked per day (information from table 2). The mean for cigarette prices in all observations is 60.3 cents per pack with a standard deviation of 4.73 cents and minimum value of 44 cents and maximum value of 70.13 cents. The mean for education is 12.47 years of education with a standard deviation of 3.06 years with a minimum value of 6 years and maximum of 18 years. The mean of age is 41.23 years with a standard deviation of 17.03 years with a minimum value of 17 and a maximum value of 88. The mean for income is 19304 US dollars per year with a standard deviation of 9142 US dollars per year with a minimum value of 500 US dollars per year and a maximum value of 30000 US dollars per year. Whereas in restaurant restriction, because its a dummy variable, we have to interpret the data slightly differently. If we multiply the mean of ‘restaurn’ which is 0.2466 with the number of observations which is 807 we get 199 which means that from the 807 people in the sample, 199 of them have state smoking restrictions in restaurants where they live. These numbers can all be found in Table 2.

The scatter plot (Figure 2) does not offer many conclusions. There does not appear to be a correlation between cigarette price and consumption. The reason for this is unknown, but it could possibly be attributed to the fact that people who don’t smoke at all are included in the data set. A change in the price of cigarettes would have no impact on these people’s tendency to smoke more or less.

The histogram (Figure 1) shows that the majority of cigarette consumption lies around 0. Specifically, this is roughly 22% of the population surveyed. It is interesting to see spikes in consumption around 20, 30, and 40 cigarettes a day; this could likely be due to the participants in this data rounding their consumption to landmark numbers (multiples of 10), but we can't say for sure why this is. Regardless, the histogram shows that the vast majority of observations in the data set smoke 20 or fewer cigarettes a day.

IV. Empirical Strategy

A model for a multiple regression model:

$$Yi = ß_0 + ß_1X_1 + ß_2X_2 + \ldots + ß_nX_n + \varepsilon_i$$

Dependent variable/Primary variable (Yi):

Yi : 'cigs'

Independent variables/Secondary variables (Xi):

X_1 : 'cigpric'

X_2 : 'educ'

X_3 : 'age'

X_4 : 'income'

X_5 : 'restaurn'

X_6 : 'hsgrad'

In our study, the regression model looks like:

$$Yi = ß_0 + ß_1X_1 + ß_2X_2 + ß_3X_3 + ß_4X_4 + ß_5X_5 + ß_6X_6 + \varepsilon_i$$

One of the variables of interest in our analysis is income. It is to be expected that people with lower incomes would not be able to afford very expensive cigarettes. However, from real

world experience we have noticed that people who smoke a lot do not make a lot of money. This is where the dummy variable that we created "hsgrad" becomes important. Another societal trend is that less educated people (who also typically make less money) tend to smoke more. Therefore, it could be possible that being uneducated about the health risks of cigarettes leads to habitual smoking even when prices are high. The addictive power of nicotine, once chemically present in a smoker's biological processes, could overcome the desire to save money. In contrast, someone who is well educated may choose not to smoke in the first place despite the fact that they could most likely afford the cigarettes better than an uneducated person. Another variable of interest related to policy formation is "restaurn." Although restaurant smoking restrictions were not discussed in our literature resources, these policies could very well influence people to not smoke as frequently. Smoking is often a social activity, and lots of people socialize by going out for meals. Deferring people from smoking as a social activity would likely have an impact on them become habitual smokers, ultimately decreasing cigarette consumption.

V. Results and Analysis

Our team has collected an array of regression models by comparing different variables to regress in Stata. From table 3, we get:

$$cigs_i = 10.67457 + (-0.0329696cigpric_i) + \varepsilon_i$$

From table 4, we get:

$$cigs_i = 9.650224 + (-1.331731hsgrad_i) + \varepsilon_i$$

From table 5 we get:

$$cigs_i = 13.66701 + (-0.3764399educ_i) + (-.0419294age_i)$$

Form Table 6 we get:

$cigs_i = 14.74315 + (-0.320155cigpric_i) +(-0.03764399age_i) + (-0.2607227educ_i) + (0.0001178income_i) + \varepsilon_i$

From Table 7 we get:

$cigs_i = 12.51743 + (0.0052361cigpric_i) +(-0.2705967age_i) + (-0.0460526educ_i) + (0.0001304income_i) + (-2.997874restaurn) + (-12.51743hsgrad) + \varepsilon_i$

According to the regression models, two variables that have a very significant effect on the consumption of cigarettes are education and restaurant restrictions. When looking at the regression in table 5, our education coefficient (-.3764399) suggests that for every year a person becomes more educated, cigarette consumption decreases by 0.37 cigarettes. Additionally, the restaurant regulations coefficient of -2.997874 in Table 7 suggests that when restaurant restrictions are put in place, cigarette consumption drops significantly. This shows just how effective restaurant restrictions and non-smoking laws are in reducing cigarette consumption. According to the regression model equation in Table 3, a 1 point increase in cigpric decreases consumption by -0.0329696 cigarettes per day. This result is practically significant, because it could be expected that higher cigarette prices influence people to purchase fewer of them. The results from Table 6 also tell us that increasing income (X_4) by one point leads to a .0001178 increase in cigarette consumption. Increasing the income of an a person addicted to cigarettes is likely to increase their consumption, because the person has more funds to support their habit. Finally, age (X_3) was also an interesting variable to observe. According to Table 5, a 1 point increase in age led to a -.0419294 change in cigarette consumption in cigarettes consumption. This could be because people becomes more educated as they get older, and choose to smoke less because they know the devastating effects smoking can have on the body. The younger observations could easily be more swayed by popular culture and the “cool vibe” that smoking

can sometimes be associated with in younger members of society. According to Table 7, the regression model that includes the dummy variables and all the other variables shows a positive coefficient for cigarette consumption and cigarette price (0.0052361), implying that higher prices in fact lead to more cigarettes consumed. This is a significant change, because in the simple regression from Table 3 this coefficient was negative. Adding more variables likely skews the results significantly, which could be the reason for this change.

From our confidence intervals in the regression for cigarette consumption against state cigarette price, education, age, income, and state smoking restrictions in restaurants, we found that there are certain variables that can be considered statistically insignificant and significant at the 95% confidence interval. Cigarette price surprisingly does not have statistical significance in its correlation with cigarette consumption. This is due to the fact that the confidence interval for the coefficient of cigarette price includes 0 in the 95% confidence level. One other variable that doesn't have statistical significance in their correlation with cigarette consumption is age. On the other hand there are 3 variables that hold statistical significance with its correlation with cigarette consumption. These variables are: education, income, and restaurant restrictions on smoking. However, the statistical significance for income isn't strong because its confidence interval at the 95% level is extremely close to 0 so its correlation with cigarette consumption is almost non-existent. Whereas the other two variables show a pretty good correlation with cigarette consumption.

VI. Conclusion

It's hard to make finite conclusions from only looking at a select group of variables, but nonetheless some patterns can be observed. The importance of restaurant restrictions and education tops all the other variables. So, these two factors could be easily used by policymakers

and institutions seeking to influence cigarette consumption. For example, more anti-smoking policies in restaurants would have a very significant effect on overall consumption. Having policymakers who put it in their agenda to enforce restaurant and public smoking laws is important for decreasing smoking. Another interesting trend is that income and price do not seem to have a huge impact on cigarette consumption. This could be because once the addiction has set in, many people do not care about how much the cigarettes cost anymore; their demand is very inelastic, and they will purchase them anyway. As long as the person can afford to smoke they will continue to do so, because it is extremely hard to quit. The literature reviewed on this topic implied that for college students, increasing the price did have a very negative impact on cigarette consumption. Although this is not exactly what our data found, it is important to note that we did not find the specific correlation between college students, cigarette price and cigarette consumption. While it makes sense that most addicted people will continue to buy cigarettes when prices go up, college students are a different demographic because they are also some of the most educated people in the population. According to our data, more years of education decreases cigarette consumption. This data is also consistent with the second journal article that details how higher levels of education go along with lower levels of smoking. However we found as education increases in non high school graduates, cigarette consumption increases. This suggests that individuals who aren't able to graduate high school may have more exposure to cigarettes and are less educated about the health risks. One suggestion for further research could be geographic area. Personal experience tells us that smoking is much more common in certain high schools in specific parts of the country, such as California. While this is a cultural trend that may not have a specific reason besides tradition, it would be interesting to see if this would have an impact on results. It would be interesting to see how the increase in

cigarette prices changes consumption in places such as California versus Massachusetts. This type of study is a suggested topic for future research.

VII. Graphs, Tables, Charts, and Diagrams

Table 1

```
ci means cigs if hsgrad==1
```

Variable	Obs	Mean	Std. Err.	[95% Conf.	Interval]
cigs	584	8.318493	.570426	7.198153	9.438833

```
ci means cigs if hsgrad==0
```

Variable	Obs	Mean	Std. Err.	[95% Conf.	Interval]
cigs	223	9.650224	.9065172	7.863744	11.4367

Table 2

Variable	Obs	Mean	Std. Err.	[95% Conf.	Interval]
cigs	807	8.686493	.4830202	7.738367	9.634619
cigpric	807	60.30041	.166802	59.97299	60.62783
educ	807	12.47088	.1076172	12.25964	12.68212
age	807	41.23792	.5993888	40.06137	42.41447
income	807	19304.83	321.8474	18673.07	19936.59
restaurn	807	.2465923	.0151823	.2167908	.2763938

Variable	Obs	Mean	Std. Dev.	Min	Max
cigs	807	8.686493	13.72152	0	80
cigpric	807	60.30041	4.738469	44.004	70.129
educ	807	12.47088	3.057161	6	18
age	807	41.23792	17.02729	17	88
income	807	19304.83	9142.958	500	30000
restaurn	807	.2465923	.4312946	0	1

Table 3

Source	SS	df	MS			
				Number of obs	=	807
				F(1, 805)	=	0.10
Model	19.6715251	1	19.6715251	Prob > F	=	0.7467
Residual	151734.011	805	188.489455	R-squared	=	0.0001
				Adj R-squared	=	-0.0011
Total	151753.683	806	188.280003	Root MSE	=	13.729

cigs	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
cigpric	-.0329696	.1020559	-0.32	0.747	-.2332966	.1673574
_cons	10.67457	6.172959	1.73	0.084	-1.442422	22.79157

Table 4

Source	SS	df	MS			
				Number of obs	=	807
				F(1, 805)	=	1.52
Model	286.205013	1	286.205013	Prob > F	=	0.2178
Residual	151467.478	805	188.158357	R-squared	=	0.0019
				Adj R-squared	=	0.0006
Total	151753.683	806	188.280003	Root MSE	=	13.717

cigs	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
hsgrad	-1.331731	1.079791	-1.23	0.218	-3.451269	.787807
_cons	9.650224	.9185638	10.51	0.000	7.847161	11.45329

Table 5

Source	SS	df	MS			
				Number of obs	=	807
				F(2, 804)	=	2.02
Model	757.250609	2	378.625305	Prob > F	=	0.1339
Residual	150996.432	804	187.806508	R-squared	=	0.0050
				Adj R-squared	=	0.0025
Total	151753.683	806	188.280003	Root MSE	=	13.704

cigs	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
age	-.0419294	.0288231	-1.45	0.146	-.0985068	.0146481
educ	-.2607227	.1605345	-1.62	0.105	-.5758389	.0543934
_cons	13.66701	2.552046	5.36	0.000	8.657554	18.67647

Table 6

Source	SS	df	MS			
				Number of obs	=	807
				F(4, 802)	=	2.13
Model	1596.4945	4	399.123624	Prob > F	=	0.0751
Residual	150157.188	802	187.228414	R-squared	=	0.0105
				Adj R-squared	=	0.0056
Total	151753.683	806	188.280003	Root MSE	=	13.683

cigs	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
cigpric	-.0320155	.101909	-0.31	0.753	-.2320554	.1680244
educ	-.3764399	.1697686	-2.22	0.027	-.7096832	-.0431966
age	-.0413708	.0287973	-1.44	0.151	-.0978977	.0151561
income	.0001178	.000056	2.10	0.036	7.93e-06	.0002277
_cons	14.74315	6.542677	2.25	0.025	1.900359	27.58594

Table 7

Source	SS	df	MS			
				Number of obs	=	807
				F(6, 800)	=	2.64
Model	2944.13399	6	490.688999	Prob > F	=	0.0154
Residual	148809.549	800	186.011936	R-squared	=	0.0194
				Adj R-squared	=	0.0120
Total	151753.683	806	188.280003	Root MSE	=	13.639

cigs	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
cigpric	.0052361	.102532	0.05	0.959	-.1960274	.2064996
educ	-.2705967	.2459131	-1.10	0.271	-.7533078	.2121144
age	-.0460526	.028989	-1.59	0.113	-.1029562	.010851
income	.0001304	.000056	2.33	0.020	.0000205	.0002404
restaurn	-2.997874	1.131855	-2.65	0.008	-5.21963	-.7761171
hsgrad	-.9008697	1.649268	-0.55	0.585	-4.138273	2.336534
_cons	12.51743	6.618228	1.89	0.059	-.4737129	25.50857

Figure 1

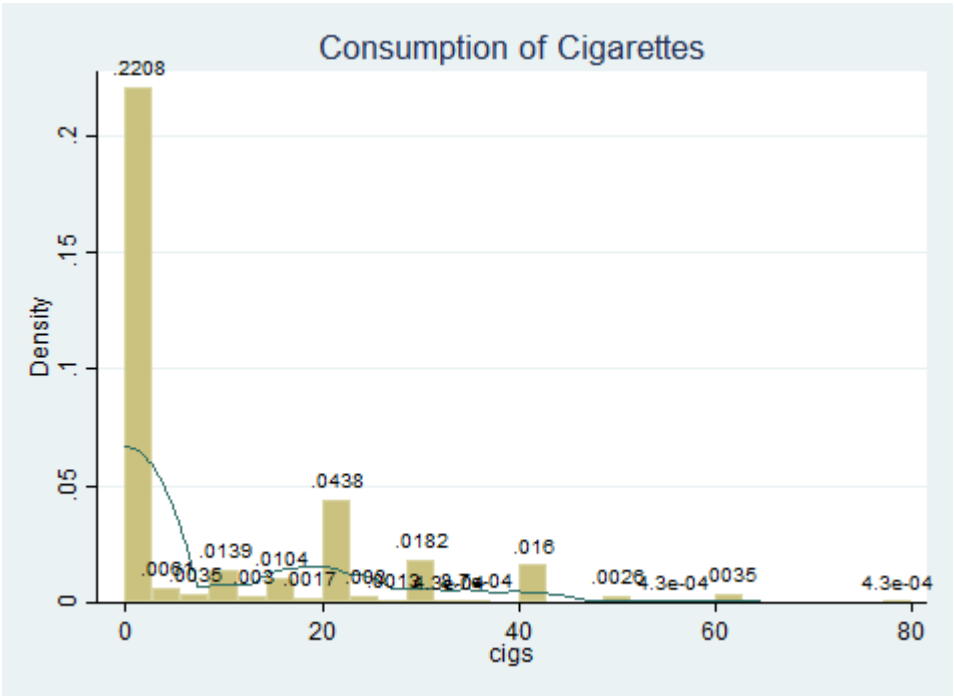
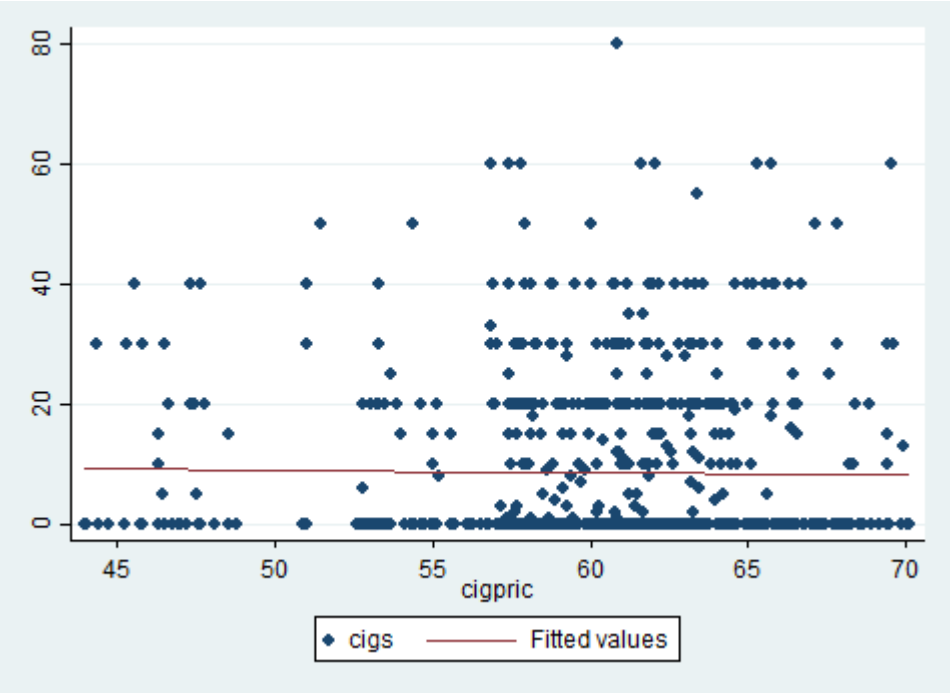


Figure 2



VIII. References

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